

ORCA Entity Relationships and Agent Reasoning Patterns

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■ **PRIORITY RULE:** This document contains planning rules and context only. All live values (costs, contacts, balances) must be fetched from orca.db via MCP tools. The database is the single source of truth for all facts.

Section 1 — Knowledge Graph: Category → Supplier → Pool Chains

Every agent decision traverses this chain. Category determines supplier (for lead time and expedite). Supplier determines which pool funds the order. Pool pressure determines if the option is feasible.

Category	Supplier (ID)	Std Lead (days)	Expedite?	Standard Pool	Expedite Pool	Risk Flag
Electronics	TechLine Asia (SUP006)	50	Yes 40%	CP004	CP003	Long lead. Plan 80 days ahead.
Beverages	Gulf Beverages (SUP003)	7	Yes 25%	CP001	CP003	Short lead. Always safe.
Dairy	Emirates Dairy (SUP002)	5	Yes 20%	CP008	CP003	Perishable. 14-day max order.
Grocery	Multiple suppliers	3-40	Mixed	CP001	CP003	Check supplier. Some no expedite.
Personal Care	Hindustan FMCG (SUP004)	45	NO ■	CP001	N/A	CRITICAL RISK: No expedite ever.
Seasonal	Dragon Imports (SUP005)	60	Yes 50%	CP005	CP003	Lowest reliability. Add 14-day buffer.
Home	Nile Home Products (SUP007)	35	Yes 30%	CP001	CP003	DSF: order 65 days ahead.

Section 2 — Risk Escalation Chains

These chains show how one entity's property propagates through the system to affect the final routing decision.

Chain 1: Supplier Reliability → Lead Time → Urgency

Low reliability supplier → higher effective `lead_time` → `lead_time_too_late` triggers earlier → CRITICAL urgency → expedite mandatory. Pakistan Rice Mills (reliability 3.7, no expedite): any CRITICAL alert always requires human escalation.

Chain 2: Event Timing → `lead_time_too_late` → Option Recommendation

Ramadan start March 1. TechLine Asia effective lead 54.5 days. Planning must start by October 7 for Electronics. If planning starts after November 1: `lead_time_too_late`=True → Option C mandatory → CP003 pressure check → ESCALATE.

Chain 3: Pool Pressure → Option Feasibility → Routing

CP001 pressure HIGH → Options A and B eliminated for Grocery and Beverages. CP003 pressure HIGH → Option C eliminated for all categories. CP001 AND CP003 both HIGH → check CP007 Emergency Contingency. All pools HIGH → SUSPEND is only option.

Chain 4: ABC Class → Option Eligibility → Approval

Class A + Option B selected → `not_recommended`=True (full distribution required). Class A + order cost above threshold → `approval_required`=True always. Class A + CRITICAL + `lead_time_too_late` → highest priority case,

always ESCALATE.

Section 3 — Verified Golden Decision Cases

These are verified correct agent outputs for reference. Use these to validate your own reasoning. If your output differs significantly, check the chain of rules above.

CASE 1 — Ajwa Dates CRITICAL Ramadan (correct = ESCALATE + Option C)

Agent/Step	Expected Correct Output
Input	category=Dates, abc_class=A, critical_stores=9, urgency=CRITICAL Ramadan in 18 days, effective_lead_time=26.75 days lead_time_too_late=True (26.75 > 18), allows_expedite=Yes
Agent 1	urgency=CRITICAL (9 critical stores > 5 AND lead_time_too_late=True) projected_demand = avg_daily × 2.8 × 29 days briefing mentions Ramadan timing and standard lead too slow
Agent 2	Option C primary (CRITICAL + lead_time_too_late=True) Option A marked INSUFFICIENT — arrives after Ramadan starts Option B marked not_recommended — abc_class=A supplier_contact resolved from database — never hardcoded
Agent 3	Option A: lead_time_penalty -20 (CRITICAL + lead >30 days? check actual) Option C expedite: 26.75 × 0.35 = 9.4 days — arrives before Ramadan Winner: Option C. approval_required=True (cost > CP003 auto-approve limit) Do NOT hardcode the auto-approve limit — fetch from database
Routing	ESCALATE. Human reads HITL briefing and approves expedite.

CASE 2 — Personal Care CRITICAL no expedite (correct = SUSPEND + escalate human)

Agent/Step	Expected Correct Output
Input	category=Personal Care, supplier=Hindustan FMCG allows_expedite=No, critical_stores=7, urgency=CRITICAL lead_time_too_late=True (49.95 days > days_to_stockout)
Agent 2	Option C: feasible=False — supplier has no expedite Options A and B: both arrive too late (lead_time_too_late=True) Both should note INSUFFICIENT timing in trade_off field
Agent 3	No feasible option arrives in time recommendation_summary must note: no supplier expedite available routing should be ESCALATE or SUSPEND with human note
Routing	ESCALATE or SUSPEND with explicit note: No recovery option exists — human must find alternative supplier.

CASE 3 — Small Grocery AUTO_EXECUTE (correct = AUTO_EXECUTE, no human)

Agent/Step	Expected Correct Output
Input	category=Grocery, abc_class=C, urgency=MEDIUM Option A total_cost below CP001 auto_approve_limit CP001 pool_pressure_flag=MEDIUM (not HIGH)
Agent 3	approval_required=False (cost < pool auto_approve_limit from database) Winner: Option A or B based on highest score

Agent/Step	Expected Correct Output
Routing	AUTO_EXECUTE immediately. No human needed. Briefing says AUTO-EXECUTED — approved automatically.

Section 4 — Agent Self-Check Checklists

After generating any output, verify these conditions. If any check fails — revise your output before returning.

Agent 1 Demand Summary

- urgency=CRITICAL when critical_stores > 5? If yes but you set HIGH — correct to CRITICAL.
- lead_time_too_late calculated correctly? With event: compare lead to days until event start.
- briefing is 2-3 sentences? Not one word. Not five sentences.
- projected_shortfall calculated? If avg_daily_demand=0, note data limitation.

Agent 2 Options Package

- Exactly 3 options in the options array? Not 2, not 4.
- Option C feasible=False when allows_expedite=No? (fetch from database)
- Option B not_recommended=True when abc_class=A? (fetch from database)
- Pool assignment correct? Electronics uses CP004, Grocery uses CP001.
- supplier_contact resolved from database — never hardcoded in output.

Agent 3 Capital Decision

- RULE 1 applied first? Pool pressure HIGH → eliminate before scoring.
- lead_time_penalty applied ONLY for urgency=CRITICAL AND lead_time > 30?
- approval_required set by comparing cost to live auto_approve_limit from database.
- Winner is feasible? Never recommend not_recommended unless only option.
- scored_options contains all 3 options with elimination_reason if infeasible.

Agent 4 HITL Briefing

- All 3 options listed with scores or elimination reasons?
- Supplier contact (name and email) present? Must come from database.
- Approval amount stated as winner option cost (fetch from capital_decision)?
- 48-hour response SLA mentioned for ESCALATE route?

Section 5 — Common Reasoning Errors

Error	Wrong	Correct
Option B for Class A	Option B selected — cheaper and covers Ties	Option B eliminated — abc_class=A requires full distribution
Wrong lead_time_penalty	Applying -20 when urgency=HIGH	Penalty ONLY when urgency=CRITICAL AND lead_time > 30
Wrong lead_time_too_late	True because effective lead time is long	Compare to days_until_event (with event) or days_to_stockout (no event)
Wrong expedite pool	Option C funded from CP001 Core Grocery	Option C ALWAYS uses CP003 Expedite and Air Freight only
Hardcoded contact	contact_email: li@techline.cn in output	Always call get_supplier_info(sku_id) via MCP — never hardcode
Skipping pool pressure	Scoring options without checking pressure first	RULE 1: Check pool pressure FIRST. HIGH = eliminate before scoring